Abstract

The aim of this sub-project is to develop an image processing component to support low-cost automated greenhouses. The image processing application must enable the greenhouse system to establish various aspects about the plants from the images taken. These aspects include plant health, plant growth and plant disease identification from the images of different parts of the plants in the greenhouse.

Fill out the introduction.

What is the purpose of this section?

- What is plant phenotyping?
- Discuss what can be measured by plant phenotyping?
- Why bother about plant phenotyping?

Context for plant phenotyping? It is of vital importance in plant breeding to be able to propogate desirable heritable traits from plants through controlled human action [11]. This is especially important in the context of food security where a cultivars resistant to adverse abiotic and biotic stresses affect the final overall crop yield. The literature indicates that plant phenotyping has become an effective tool in plant breeding methods. [1].

What is plant phneotyping Plant phenotyping is then the evaulation of phenotypes in plants with the main goal of producing quantative data to study the dynamic interaction between a plant and its environment [3].

Definitons 2
The phenotype can be seen as the combination of all the morphological, physiological, anatomical, chemical, developmental, and behavioral characteristics that, when put together, represent the individual organism. Phenotyping, which is the process of characterizing the phenotype [5]

Definitons 3
For the purposes of this review, we refer to phenotyping as the set of methodologies and protocols used to measure plant growth, architecture, and composition with a certain accuracy and precision at different scales of organization, from organs to canopies. The goal of modern plant phenotyping is to deliver quantitative data on the dynamic responses of plants to the environment. [3]

Definitons 4
The broad evaluation of complex plant traits like growth, development, tolerance, resistance, architecture, physiology, ecology, and yield as well as the fundamental measurement of individual quantitative parameters that serve as the foundation for more complex traits is known as plant phenotyping [10]

Definitons 5
Plant phenotyping is the comprehensive assessment of complex plant traits such as growth, development, tolerance, resistance, architecture, physiology, ecology, yield, and the basic measurement of individual quantitative parameters that form the basis for more complex traits [8]

Explanation of relevance of plant phenotyping for current study? The term phenotype was first employed by Johannsen [7]. Who stated that a phenotype is all types of organism which are idenfitied from one antoher by inspection or by some measuremment. So it is an obserable primitive trait or a complex trait of a plant that can be identified visually or measured and evaulated numerically. Therfore the data produced by plant phenotyping assist in choosing the genetic material (germplasm) that will enable plant breeders to induce traits in cultivars which are optimal for various diffrent growing environments (like water scarce environments).

Why plant phenotyping in the context of this study? Conventional plant plantphenotyping suffers from scalabality issues and is often constrained by costs, time, and human effort required. Furthermore many methods are destructive to the cultivar, making it hard to replicate, and compare measurements of phenotypes over time through the ontogensis of a cultivar. However in the past decade image processing techniques have been applied successfully to aquire phenotypical data in a non-destructive matter [2].. There exists many different image techniques that have been applied in greenhouses and in the field which allow for the assesment of resistance to disease, biomass, soli salinity, etc using various differenr sensors, and techniques.

What affects plant growth?
Plant biomass is affected by stresses and disturbances [4]. A stressor is anything which stymies the growth of plant biomass. Examples of stressors include things like water scarcity, inadequate amounts of light, and unfavourable temperatures. A disturbance is anything which causes some level of destruction to the plant. Examples of disturbances include pathogens, fires, frost, and pests.

How can we measure plant stresses and disturbances?
Stresses and disturbances of plants are induced by environmental conditions the organisms are placed in, and therefore transitively induce phenotypical traits. We can exploit this fact by measuring plant phenotypical traits, we can infer the prescence of potential stresses and disturbances on cultivars. For example thermal imaging allows to meausre the phenotype of plant temperature or plant transpiration, which in turn can give us an indication of soil salinity. We indirectly infer environmental conditions based on phenotypical traits

What physical traits of a plant can be measured phenotypically?

Conceptual basis Typically in the context of plant breeding we care about plant yield, resistant to abiotic and biotic stresses. That is in general we seek plants which are fit. In the literature of the term trait is used to describe individual scoped properties, communit scoped properties end even environmental properties. trait is any morphological, physiological or phenological feature measurable at the individual level, from the cell to the whole-organism level, without reference to the environment [15]. Plant phenotyping reflects this in its achievements and developments over the past decade. The key is environmental related performance metrics are derived from individual plant traits. So these phenotypical traits act as proxies that indirectly preditc plant fitness and performance.

Morphological traits About plant sturcture, which affects its performance and gives indication regarding its health

- Leaf width
- Number of leaves
- Leaf inclination angle
- Leaf area
- Root system architecture (e.g root diameter)
- Plant shoots
- Fruit
- Flowers

Physiological traits About how plant operates in repsonse to its growing conditions

- Stomotal conductance
- Chlorophyll content of leaf
- Plant water status
- Plant photosynthesis (metablism)
- Nutrient content

Phenological traits These are phenological traits (when certain biological events occur that depend on climates)

- When fruit or flowers are yielded
- Yield prediction

The use of plant phenotyping and detecting plant stress Plant biomass is affected by stresses and disturbances [4]. A stressor is anything which stymies the growth of plant biomass. Examples of stressors include things like water scarcity, inadequate amounts of light, and unfavourable temperatures. A disturbance is anything which causes some level of destruction to the plant. Examples of disturbances include pathogens, fires, frost, and pests

Biotic Stress + Disturbance.

- Pathogenic organisms
- Pests
- Parasitc plants

Abiotic Stress + Disturbance

- Drought
- Flood
- Salinity
- Heat Stress
- Raditation Stress, light stress

What is the purpose of this section?

- Discuss image processing in the context of plant phenotyping
- Discuss image acquistion techniques
- Discuss image processing techniques used in plant phenotyping

How does image processing relate to plant phenotyping?

Fill out this section

The conceptual pipeline for image processing techniques in plant phenotyping

This paragraph should discuss how image processing is applied. I.e it should map the process from image acquistion to image processing to data generation/analysis

Foundations of image acquistion in plant phenotyping

- measure a phenotype quantitatively through the interaction between light and plants such as reflected photons, absorbed photons, or transmitted photons. Each component of plant cells and tissues has wavelength-specific absorbance, reflectance, and transmittance properties [8]
- For example, chlorophyll absorbs photons primarily in the blue and red spectral region of visible light [8]
- Imaging at different wavelengths is used for different aspects of plant phenotyping [8].
- Imaging techniques are very helpful for detecting these properties, especially for those that cannot be seen by the naked eye [8]

Image acquistion methodologies in plant phenotyping

Thermal imaging sensors

- Conceptual basis of this technique [12]
- Have been used to measure plant water status.
- Used in Irrigation scheduling.
- By inferring transpiration rate of leaves which is governed by stomoatal conductance.
- Stomotal conductance can indicate drought, pathogens, and also soil salinity.
- Have also been used for yield predictions [6]

Hyperspectral sensors

- Based on the principles of spectroscopy. Can detric biochemical, morphological and physilogical traits of plants. Has been used to detect drought stress. Also used to detect water stress, and pathogens. Different reflectance can tell us about different things. Certain band tells us about water stress, some can tell us about clholrphyll content. [13].
- Conceptual basis and applications [**saricApplicationsHyperspectralI**]
- Crop yield
- Biotic and abiotic factors

Tomography
Fill this out

Flouresnce sensors

- Conceptual basis of this technique [14]
- Predict metabolism using photosynthesis which tells us about growth status
- Any stress or disturbance can indirectly or directly affect plant metabolism, so chlorphyll content canbe us used as a measure to detect these things like pathogens, light shortage, nitrogen deficiency, water stress etc [9]
- In optimal conditions, very little heat is emitted in photosynthesis so in certain near infrared emission should be low
- Acitinic light involved here?

RGB Camera
Fill this out

What is the purpose of this section?

- Discuss image processing in the context of plant phenotyping
- Discuss image acquistion techniques
- Discuss image processing techniques used in plant phenotyping

Fill out the conclusion section.

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